



Master Theorem and Linear Sorts, Homework 3

1 Master Theorem

Bound the following recursive functions. In each case, assume that $T(1) = T(0) = 1$.

1. $T(n) = 7T(n/3) + n^3$
2. $T(n) = T(\sqrt{n}) + 2^n$

2 β -Root Master Theorem Variation

Consider the following recursive relation:

$$T(n) = \begin{cases} \alpha T(n^{\frac{1}{\beta}}) + \log^{\gamma}(n) & \text{for } n > 1 \\ 1 & \text{else} \end{cases}$$

When $\alpha, \beta > 1$ and $\gamma \geq 0$. Bound T asymptotically tight, namely find $g : \mathbb{N} \rightarrow \mathbb{R}^+$ such that $T = \Theta(g)$. Notice that the solution depends on the relation between α, β, γ .

3 Stability

For each of the following sorts, prove or disprove whether they are stable.

1. Bubble sort.
2. Counting sort.

4 Sorts That Assume Input Properties

In any of the following sections, we make another assumption about the input. For each section, write an algorithm that uses the assumption to sort the inputs and compute its running time. You don't have to provide a correctness proof.

1. $x_1, x_2, \dots, x_n \in \mathbb{R}$ such that $\max_i \{x_i\} - \min_i \{x_i\} \leq 5n$ and $|x_i - x_j| > 1$ for any $i \neq j$.
2. $x_1, x_2, \dots, x_n \in \mathbb{R}$ such that $\max_i \{x_i\} - \min_i \{x_i\} \leq 5n$ and $|x_i - x_j| > \Delta$ for any $i \neq j$, but it is unknown what Δ is. The running time should depend on Δ .

5 Camels

Bob is willing to send Alice camels, $c_1, c_2, \dots, c_n$ in a specific order. Unfortunately for him, Eve doesn't like him. Hence, she decides to spoil their order by stacking camels as follows: Eve chooses camel c_i and a position j , and then moves the i th camel c_i between c_j and c_{j+1} . She can move as many camels as she wishes. Alice, known for her strictness, wouldn't accept camels standing out of order. Luckily, Bob is allowed to send the camels again in their original order for five times, and Eve cannot choose the same camel twice. In other words, if Eve chooses to move c_i in the first attempt, then she cannot move it in any of the other four attempts.

Alice records the positions of the camels every time they come. Write an algorithm that will help Alice restore the order. (Prove correctness and compute the running time.)